Department of Computer Science and Engineering (CSE)
BRAC University

Practice Problem Set 3.4

CSE251 – Electronic Devices and Circuits

DIODE RECTIFIERS

Haft Wave and Full Wave Rectifiers with and without Filter Capacitor, Average or DC Voltage, and Input Output Waveforms

Course Description, COs, and Policies



Midterm and Final Questions

For any error or issues, contact purbayan.das@bracu.ac.bd

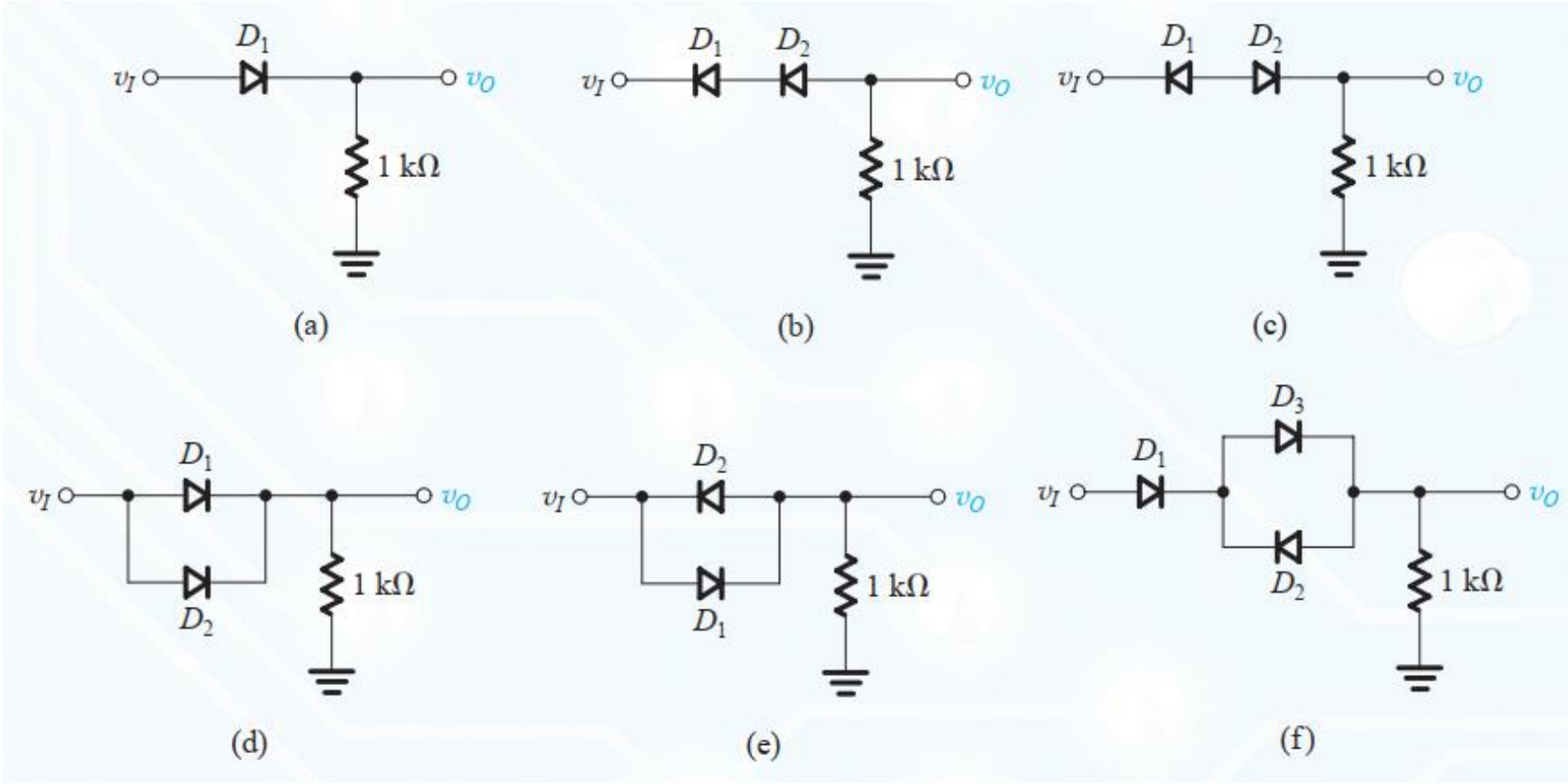
Input Voltage Parameters	Value
Peak	V_m
Frequency	f
Time period	$T = \frac{1}{f}$

Without Filter Capacitor	Output Voltage Frequency	Output Voltage Peak (V_P)	Average output voltage (V_{DC} or V_{Avg})	Ripple Voltage (V_r)	Average Output Current (I_{Avg} or I_{DC})
HW	f	$V_m - V_{D_o}$	$\frac{V_m}{\pi} - \frac{V_{D_o}}{2}$	V_P	V_{DC}/R
FW	$2f$	$V_m - 2V_{D_o}$	$\frac{2V_m}{\pi} - 2V_{D_o}$	V_P	V_{DC}/R

With Filter Capacitor	Output Voltage Frequency	Output Voltage Peak (V_P)	Average output voltage (V_{DC} or V_{Avg})	Ripple Voltage (V_r)	Average Output Current (I_{Avg} or I_{DC})
HW	f	$V_m - V_{D_o}$	$V_P - \frac{V_r}{2}$	$\frac{V_P}{fRC}$	V_{DC}/R
FW	$2f$	$V_m - 2V_{D_o}$	$V_P - \frac{V_r}{2}$	$\frac{V_P}{2fRC}$	V_{DC}/R

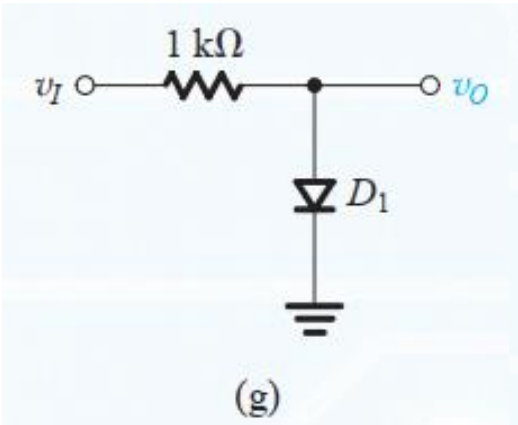
Problem 1.1

- In each of the ideal-diode circuits shown below, v_I is a $1\ kHz$, $10\ V$ peak sine wave. Sketch the waveform resulting at v_O. What are its positive and negative peak values?

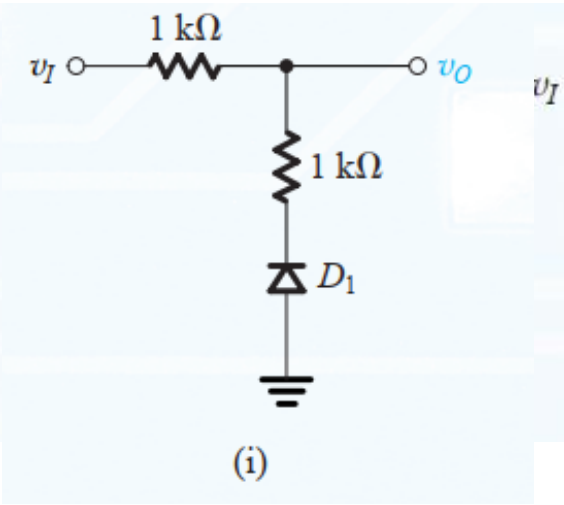


Problem 1.2

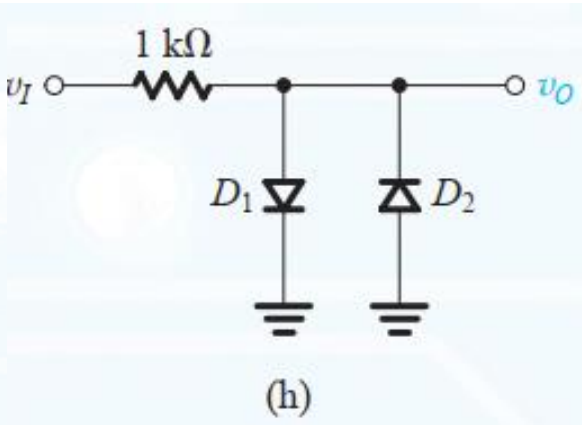
- In each of the ideal-diode circuits shown below, v_I is a $1\ kHz$, $10\ V$ peak sine wave. Sketch the waveform resulting at v_O. What are its positive and negative peak values?



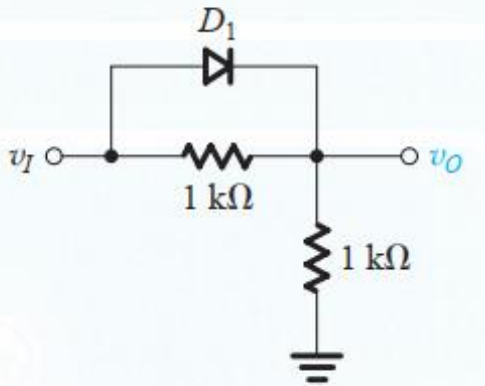
(g)



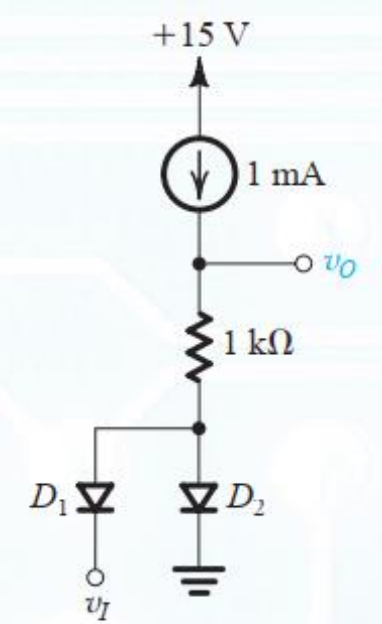
(i)



(h)



(j)



(k)

Problem 2

- The diode in the adjacent circuit has a cut-in voltage of $1\ V$.
  a) For current to flow, v_I must be greater or smaller than what voltage?

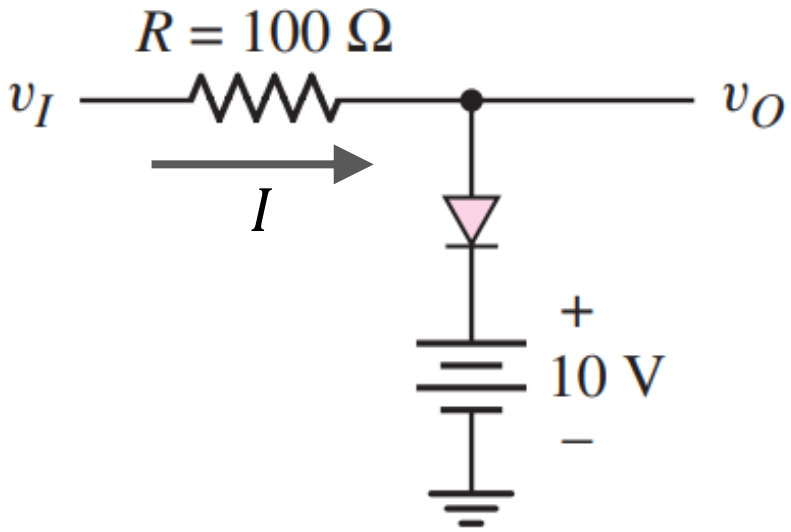
  [Hint: Assuming the diode is ON, use KCL/KVL to write an equation for the current I, then solve the inequality $I > 0$]

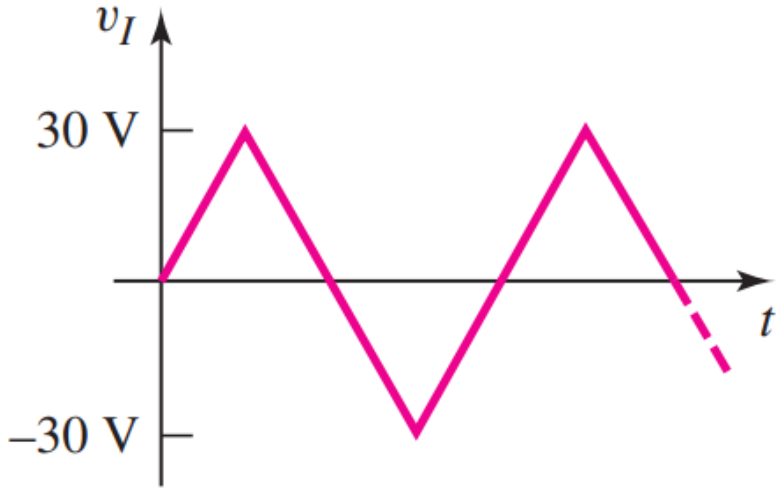
  b) If the condition in (a) is satisfied, and the diode is ON, what is the value of v_O?

  c) Write an equation for v_O in terms of v_I if the diode remains OFF.

  d) Draw the VTC (v_O vs. v_I) of the circuit.

  e) Using the answers in (b) and (c), and given that v_I is a triangular wave as shown, sketch $v_O(t)$.





Ans: $(a) > 11\ V; (b)\ 11\ V; (c)\ v_O = v_I$

Problem 3

- The diode in the adjacent circuit has a cut-in voltage of $1\ V$.

  a) For current to flow, v_I must be greater or smaller than what voltage?

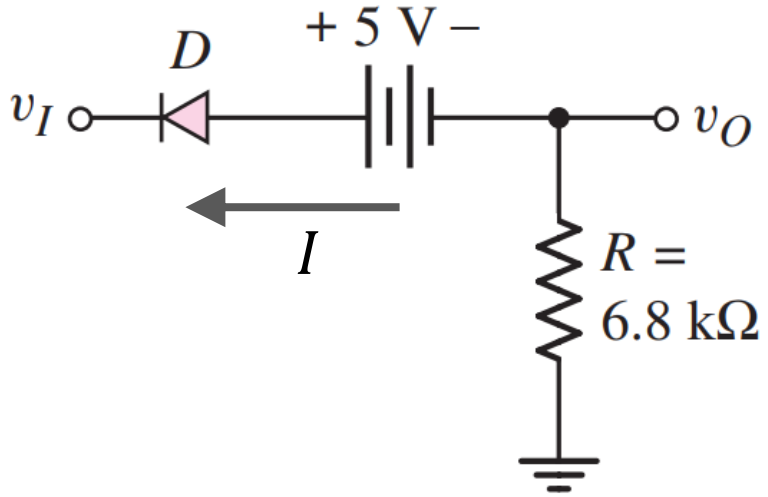
  [Hint: Assuming the diode is ON, use KCL/KVL to write an equation for the current I, then solve the inequality $I > 0$]

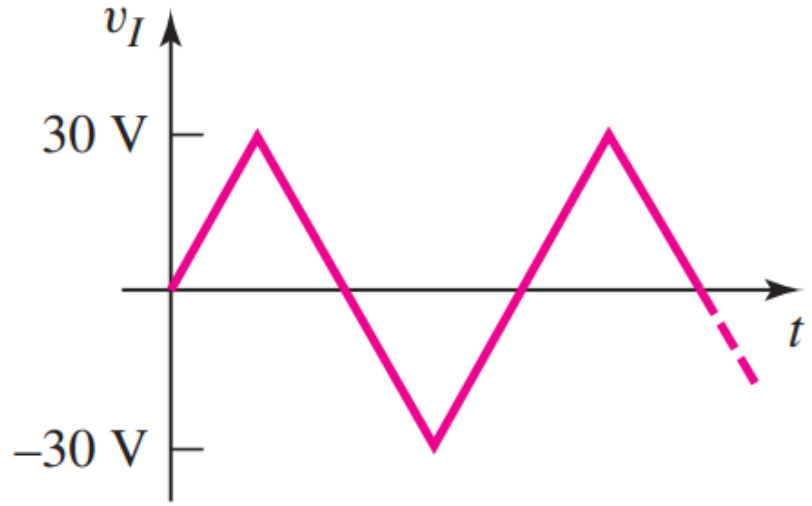
  b) If the condition in (a) is satisfied, and the diode is ON, write an equation for v_O in terms of v_I.

  c) What is the value of v_O if the diode remains OFF?

  d) Draw the VTC (v_O vs. v_I) of the circuit.

  e) Using the answers in (b) and (c), and given that v_I is a triangular wave as shown, sketch $v_O(t)$.





Ans: $(a) < 4\ V$; $(b)\ v_O = v_I - 4$; $(c)\ 0\ V$

Problem 4

- The diode in the adjacent circuit has a cut-in voltage of $1\ V$.

  a) For current to flow, v_I must be greater or smaller than what voltage?

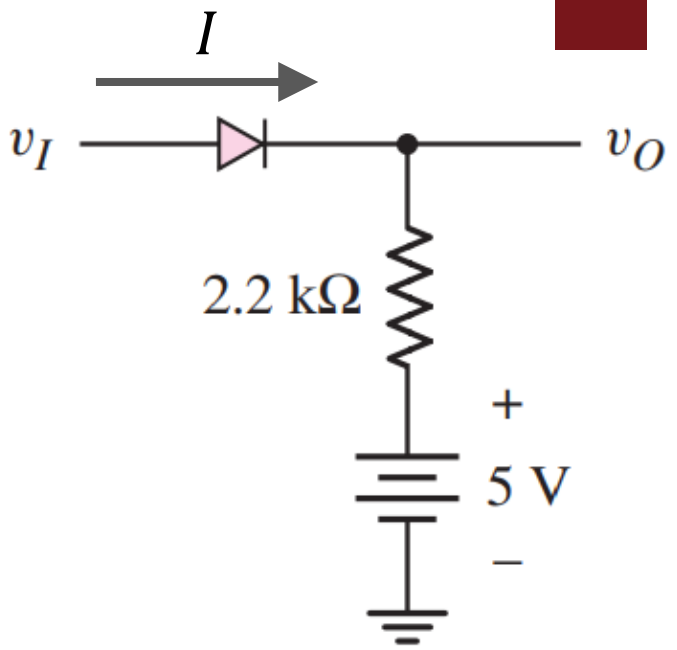
  [Hint: Assuming the diode is ON, use KCL/KVL to write an equation for the current I, then solve the inequality $I > 0$]

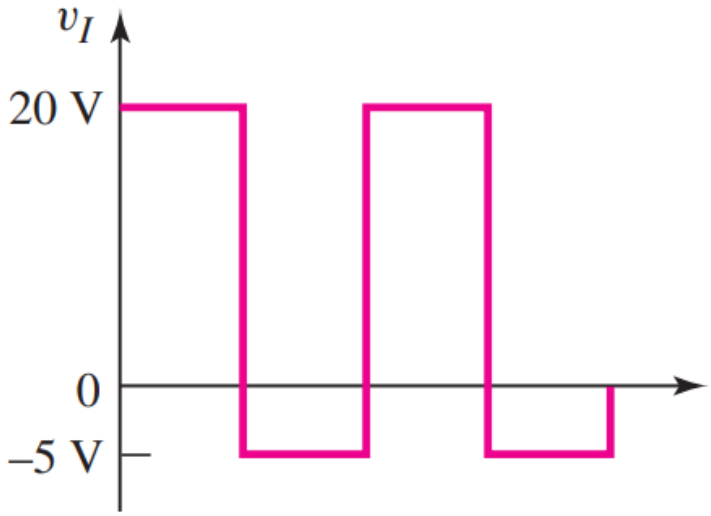
  b) If the condition in (a) is satisfied, and the diode is ON, write an equation for v_O in terms of v_I.

  c) What is the value of v_O if the diode remains OFF?

  d) Draw the VTC (v_O vs. v_I) of the circuit.

  e) Using the answers in (b) and (c), and given that v_I is a square wave as shown, sketch $v_O(t)$.





Ans: $(a) > 6\ V$; $(b)\ v_O = v_I - 1$; $(c)\ 5\ V$

Problem 5

Ans: $(a) < -3\ V$; $(b)\ v_O = v_I + 3$; $(c)\ 0\ V$

- The diode in the adjacent circuit has a cut-in voltage of $1\ V$.

  a) For current to flow, v_I must be greater or smaller than what voltage?

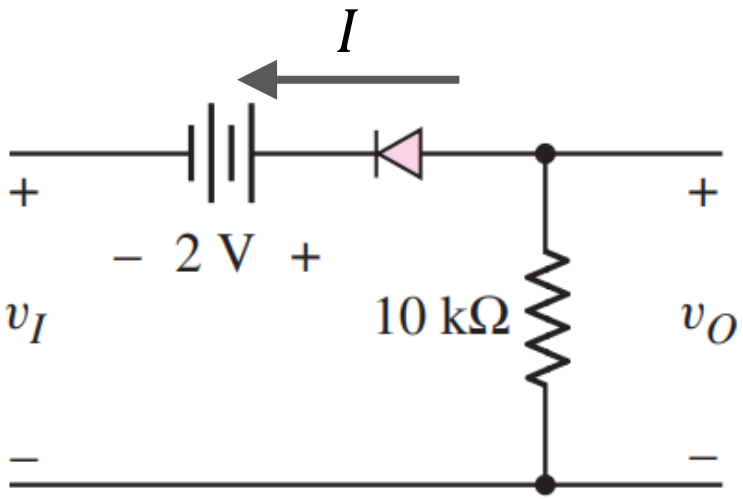
  [Hint: Assuming the diode is ON, use KCL/KVL to write an equation for the current I, then solve the inequality $I > 0$]

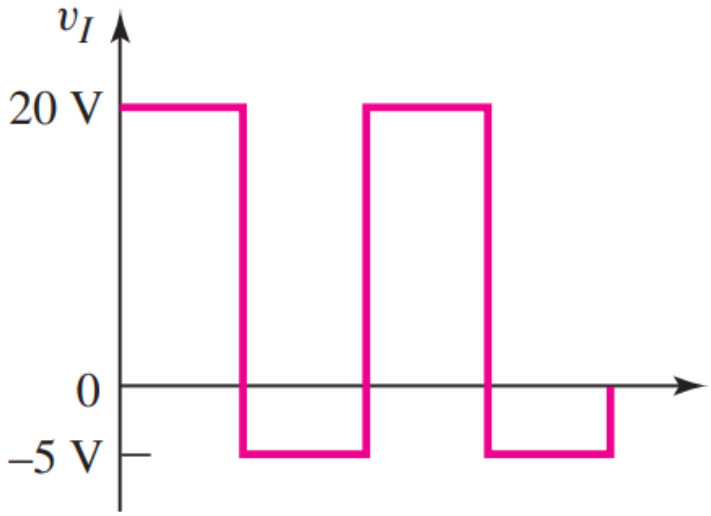
  b) If the condition in (a) is satisfied and the diode is ON, write an equation for v_O in terms of v_I.

  c) What is the value of v_O if the diode remains OFF?

  d) Draw the VTC (v_O vs. v_I) of the circuit.

  e) Using the answers in (b) and (c), and given that v_I is a square wave as shown, sketch $v_O(t)$.





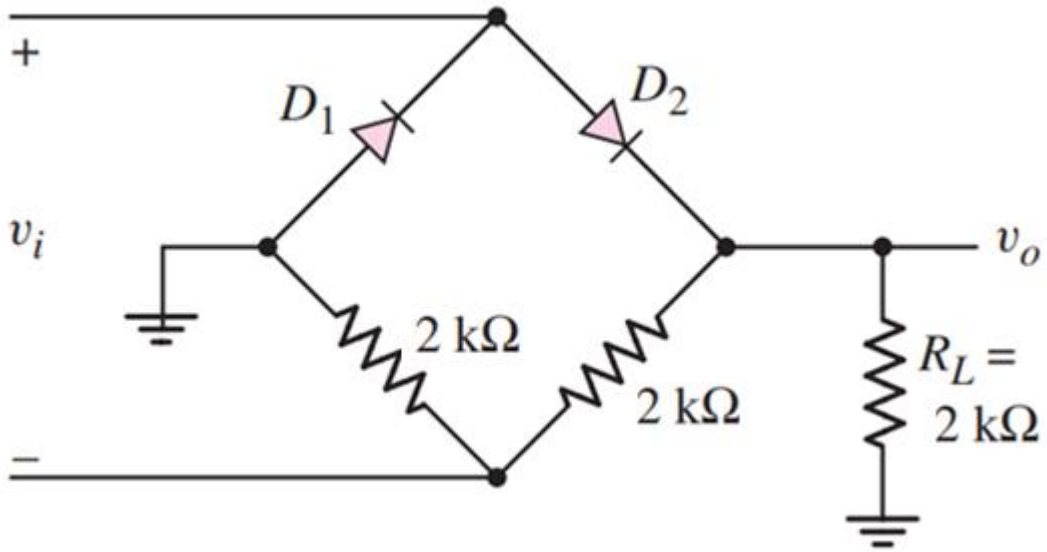
Problem 6**

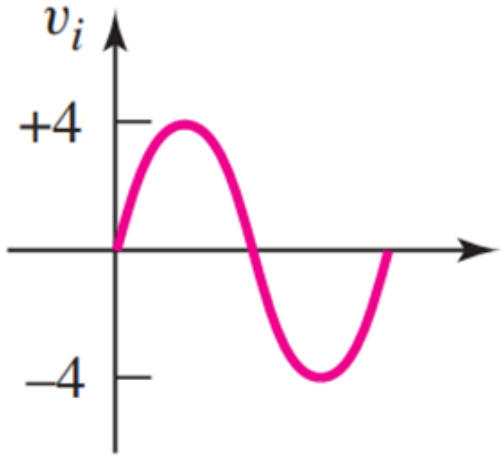
- The circuit below rectifies an ac voltage into dc. The two diodes are ideal and identical. It can be shown that during each half cycle, only one diode remains ON: D_2 in '+' half cycles and D_1 in '−'s.

  a) Derive the voltage transfer characteristics (VTC), that is, an expression relating v_O and v_i.

  [Hint: Analyze separately for '+' and '−' half cycles and you should get the same relationship.]

  a) Plot the VTC derived in (a).





Ans: $(a)\ v_o = \pm {}^{v_i}/_2$

Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

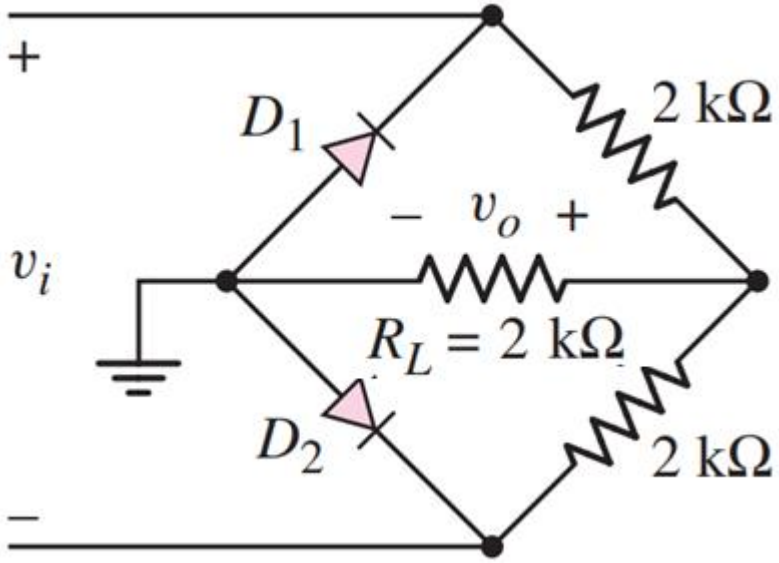
Problem 7**

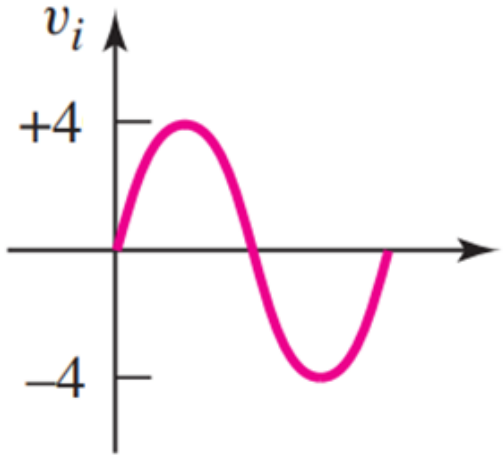
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  a) Derive the voltage transfer characteristics (VTC), that is, an expression relating v_O and v_i.

  [Hint: Analyze separately for '+' and '−' half cycles and you should get the same relationship.]

  a) Plot the VTC derived in (a).





Ans: $(a)\ v_O = \pm {}^{v_i}/_3$

Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

Problem 8

- Assuming a cut-in voltage $1\ V$ for diode(s), draw the VTC plot for
  a) Half-wave rectifier
  b) Full-wave rectifier

Problem 9

- The input voltage to a half wave rectifier is $75\sin(120\pi t)\,V$. The voltage drop across the diode is negligible compared to the input voltage. The ripple voltage is to be no more than $4\,V$. If the filter capacitor is $50\,\mu F$, determine the minimum load resistance that can be connected to the output.

Ans: $\mathbf{6.25\,k\Omega}$

Problem 10

- The output voltage of a full wave bridge rectifier has been filtered with two $2.5\ \mu F$ capacitors in parallel. The peak, average, and frequency of the output voltage are $4.3\ V$, $4\ V$, and $300\ Hz$ respectively. Determine the load resistance.

Ans: $\mathbf{4.78\ k\Omega}$

Problem 11

- Assume that the input signal to a rectifier circuit has a peak value of $12\ V$ and is at a frequency $50\ Hz$. Assume the load has a resistance of $2\ k\Omega$ and the ripple voltage is to be limited to $0.5\ V$. Determine the value of capacitance required to yield this specification for (a) half wave rectifier and (b) full wave bridge rectifier. Assume a $0.5\ V$ cut-in voltage for each diode.

Ans: **(a) $230\ \mu F$; (b) $220\ \mu F$**

Problem 12

- A $24\,V$ peak to peak $50\,Hz$ sine wave ac voltage is to be rectified with a full wave bridge rectifier circuit without filter capacitor. Determine the portion of the output voltage cycle during which each diode conducts. Also, calculate the average output voltage. Assume all the diodes are identical with a cut-in voltage $V_{D_o} = 0.6\,V$.

Ans: $\mathbf{93.62\%}$; $\boldsymbol{V_{avg} = 6.44\,V}$

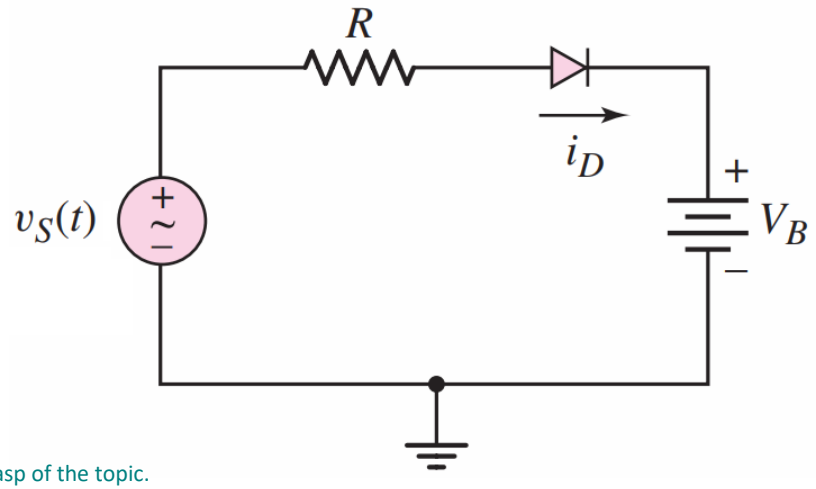
Problem 13

- A full wave bridge rectifier is to be designed to produce a peak output voltage of $12\ V$, deliver peak $120\ mA$ to the load, and produce an output with a ripple of not more than 5%. An input line voltage of $9.9\sqrt{2}\ V\ (peak)$, $2\ kHz$ is available. Specify the diode voltage and value of the capacitance you require.

Ans: $\mathbf{1\ V;\ \ 50\ \mu F}$

Problem 14

- A battery V_B is charged though a half wave rectifier circuit as shown below. The voltage V_B of the battery at the time of charging is $4.5\ V$. The resistance R is $250\ \Omega$. $v_S(t)$ is a $24\ V$ peak to peak sine wave. Cut-in voltage for the diode is $V_{D_o} = 0.6\ V$.
  I. What is the required value of $v_S(t)$ for the diode to conduct?
  II. Determine the peak diode current, maximum reverse-bias diode voltage, and the fraction of the cycle over which the diode is conducting.
  III. **Determine the average voltage of the voltage across the resistor. *[Hint: note that, the average formula we derived will not work here. You have to integrate with the exact limits.]*

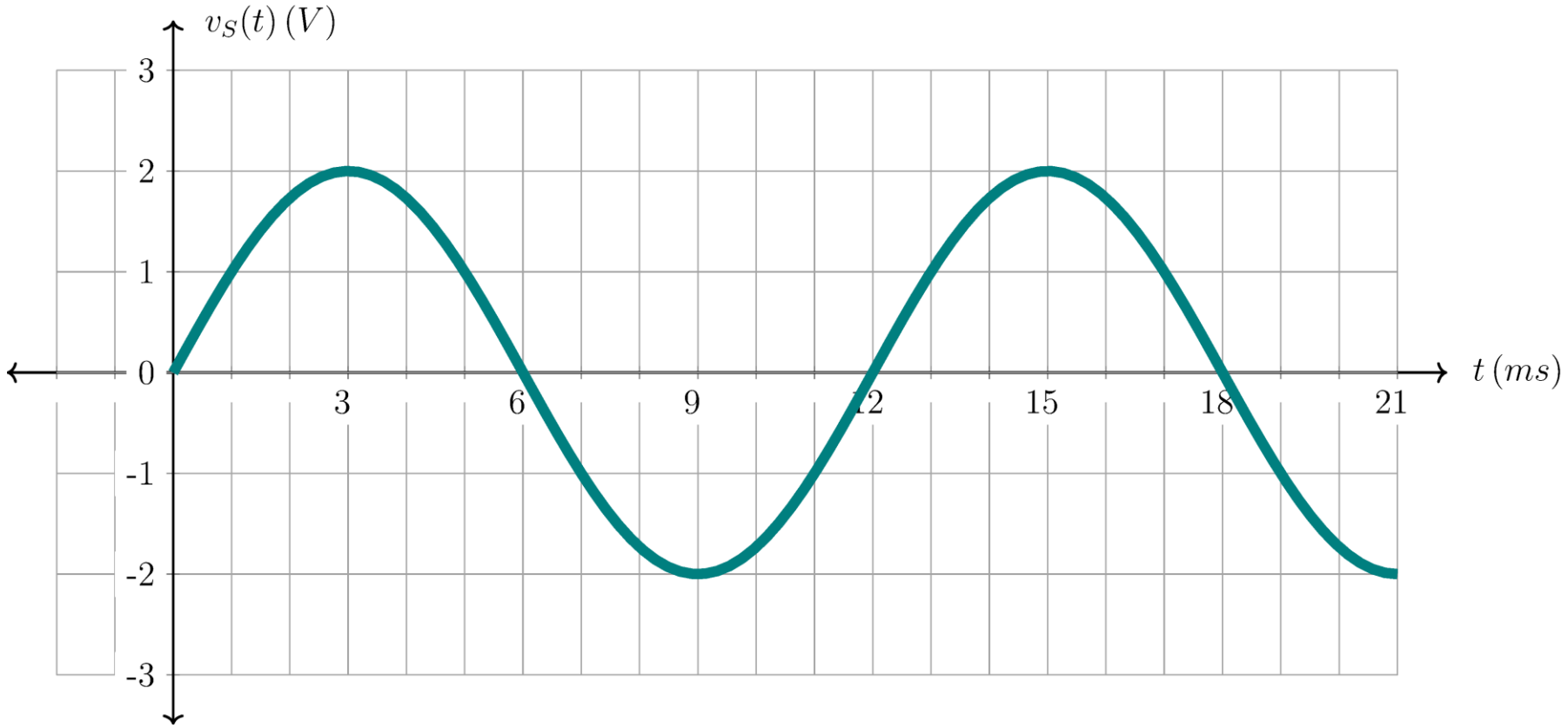


Ans: I. $\mathbf{5.1\ V}$; II. $\mathbf{27.6\ mA, 16.5\ V, 36\%}$; III. $\boldsymbol{V_{avg} = 3.47\ V}$

Note: Problems marked with an asterisk (**) are a bit more advanced for this course. However, attempting them can help you develop a stronger grasp of the topic.

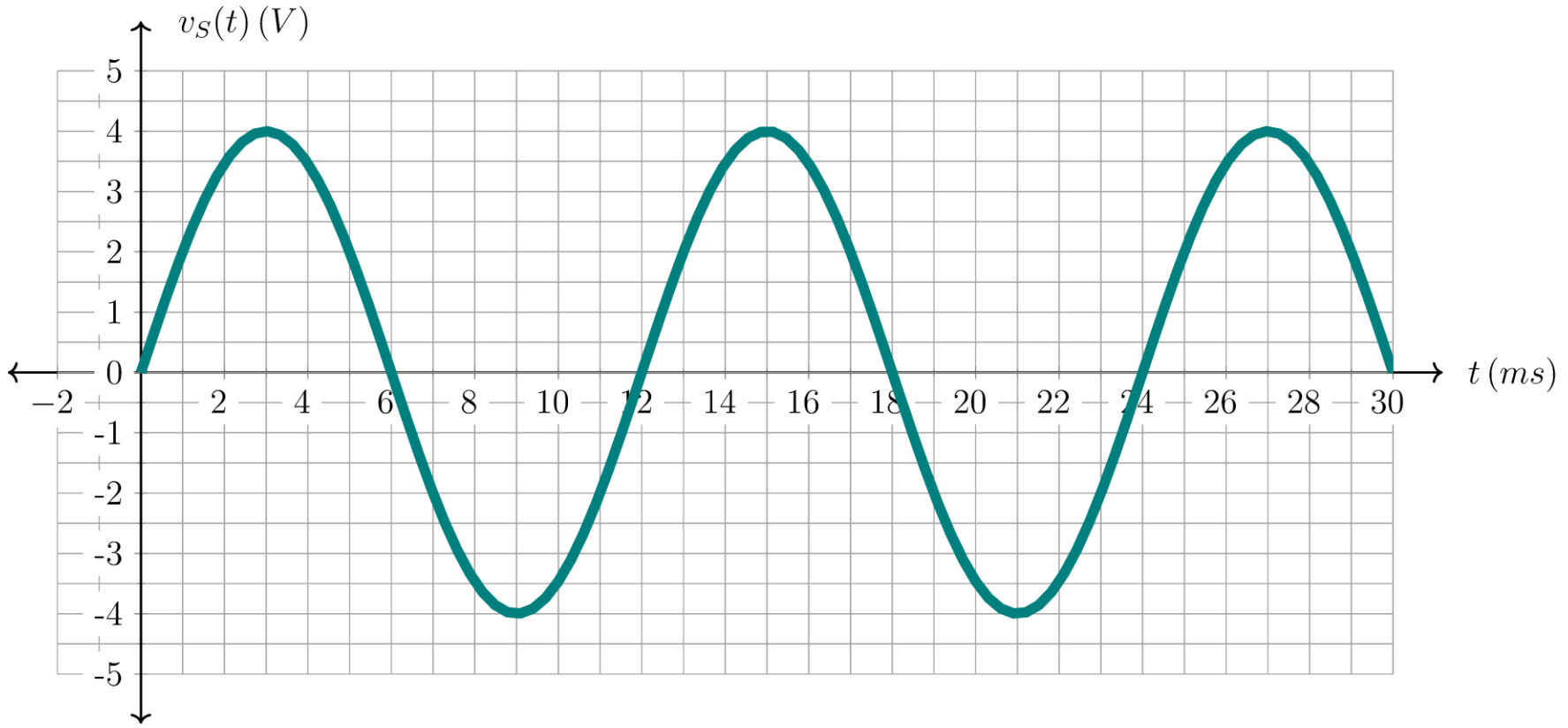
Problem 15

- The following is the input voltage to a half wave rectifier without filter capacitor. Draw the output waveform on the same grid. Assume a cut-in voltage $V_{D_o} = 1.0\ V$ for the diode.



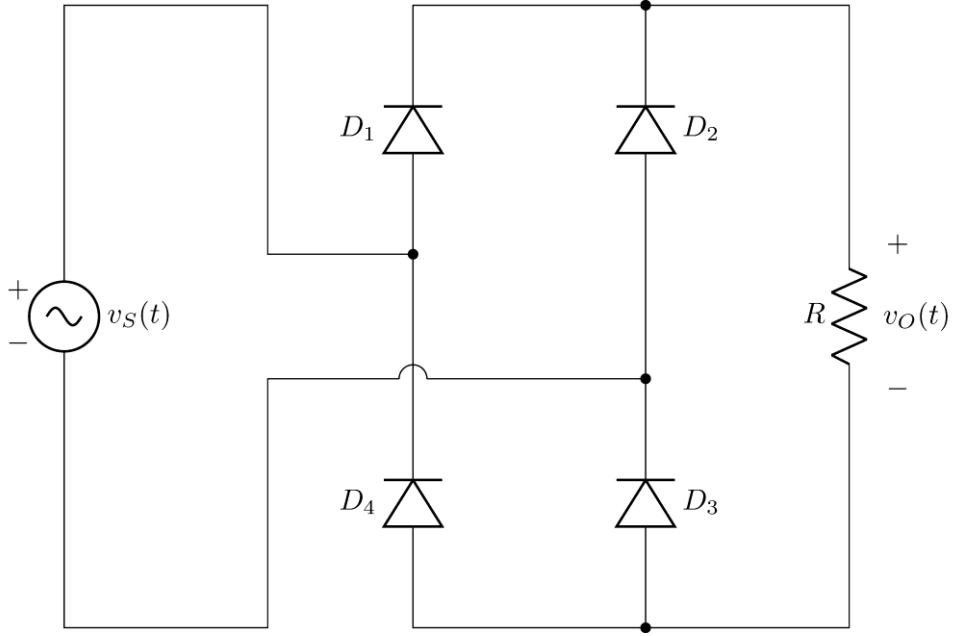
Problem 16

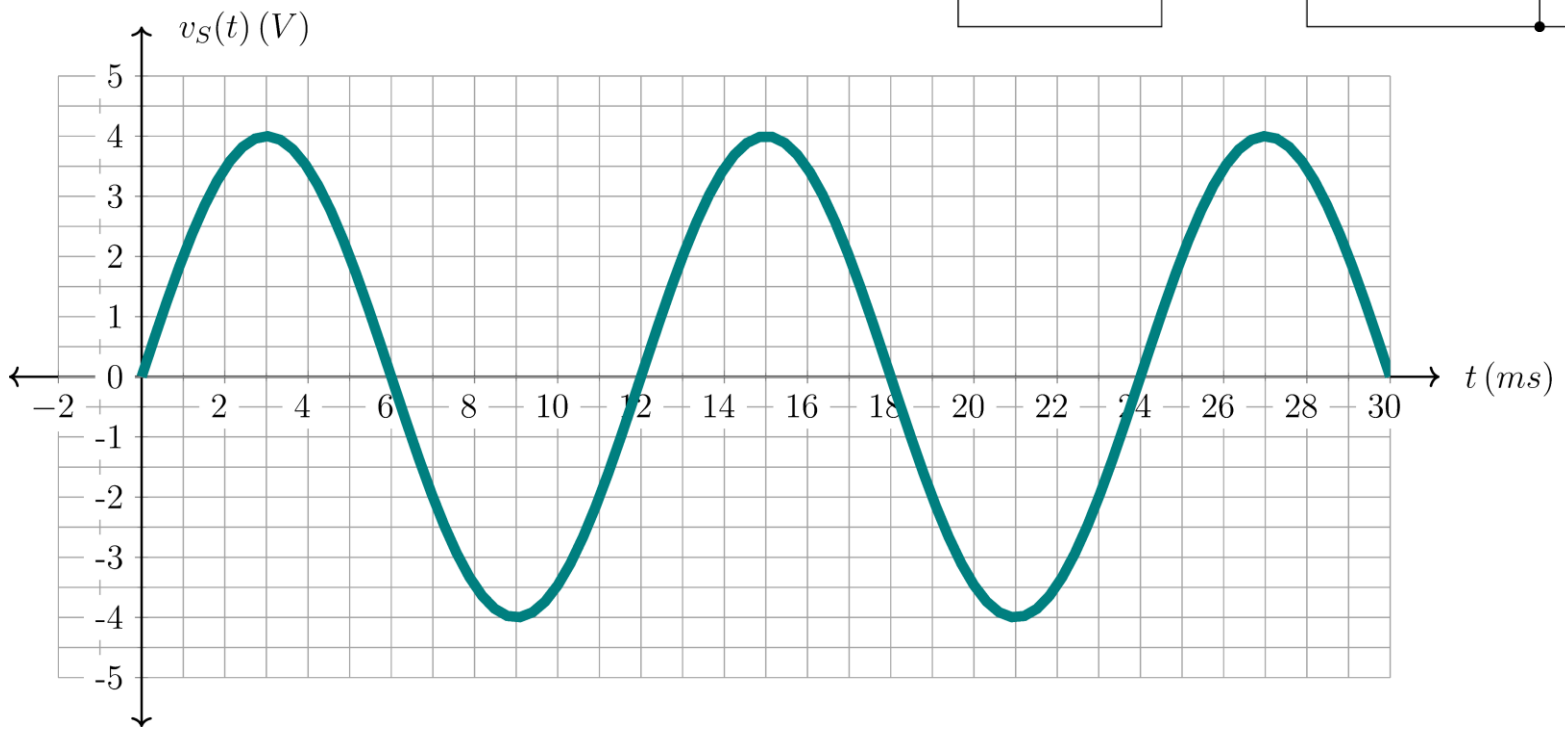
- The following is the input voltage to a full wave bridge rectifier without filter capacitor. Draw the output waveform on the same grid. Assume all the diodes are identical with a cut-in voltage $V_{D_o} = 1\ V$.



Problem 17

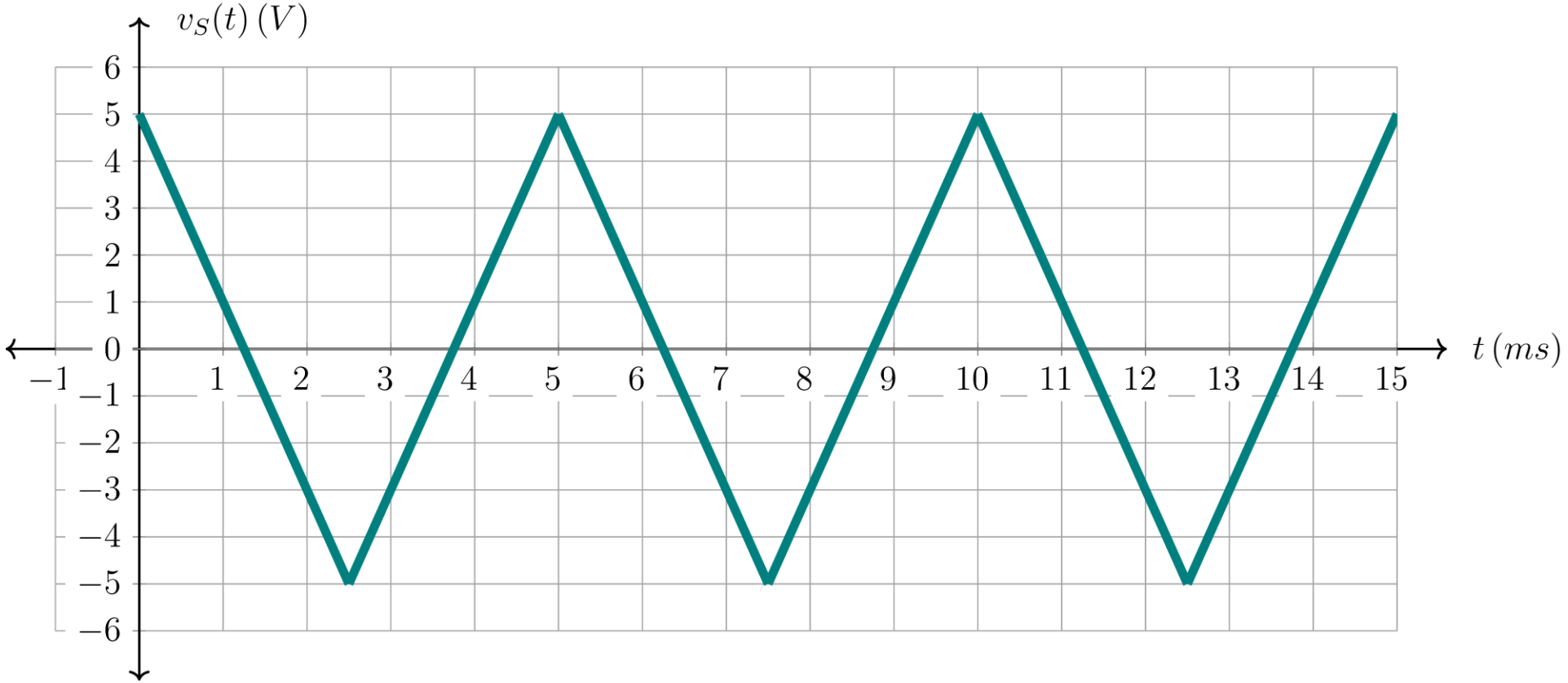
- The following is the input voltage to a full wave bridge rectifier without filter capacitor. Draw the output waveform on the same grid. Assume $V_{D_{o_1}} = 1.2\,V$, $V_{D_{o_2}} = 0.55\,V$, $V_{D_{o_3}} = 0.8\,V$, and $V_{D_{o_4}} = 0.45\,V$.





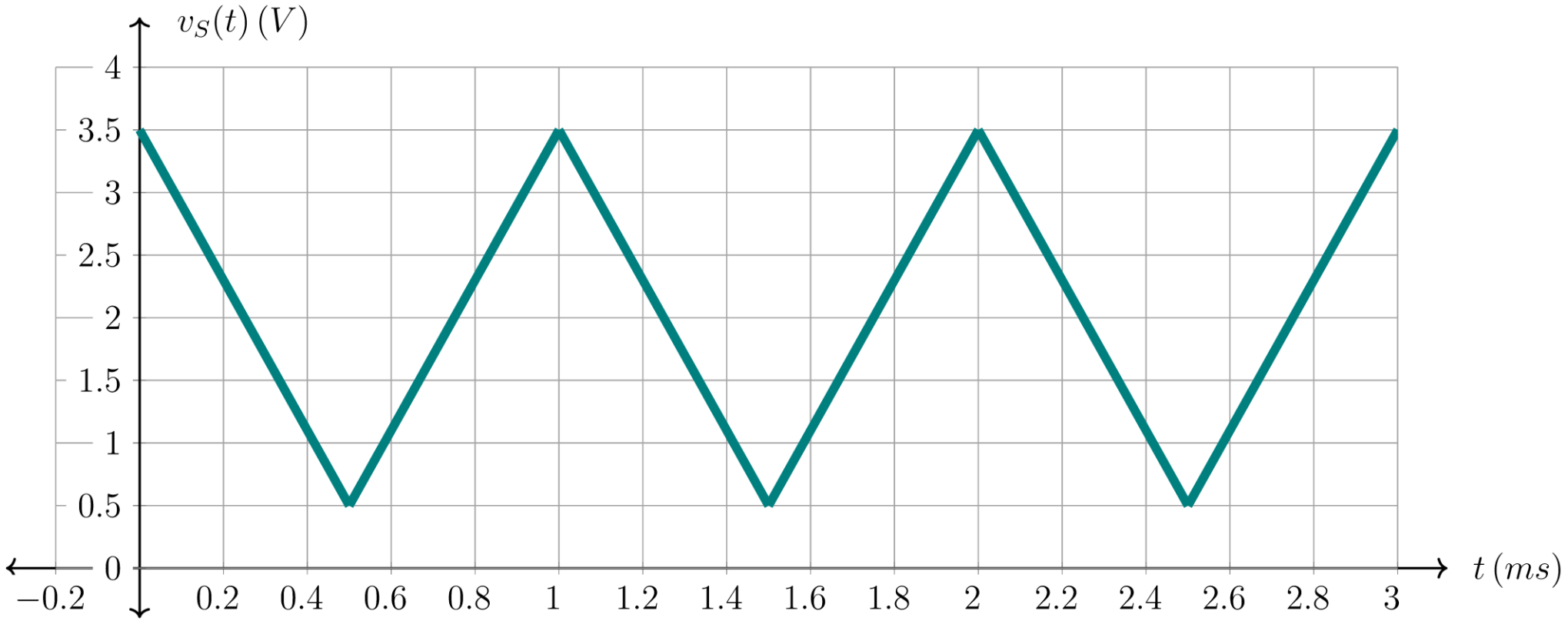
Problem 18

- Draw the output voltage waveforms if the following is the input voltage to a (i) half wave rectifier and (ii) full wave bridge rectifier, both without filter capacitor. Assume all the diodes are identical with a cut-in voltage $V_{D_o} = 1\ V$.



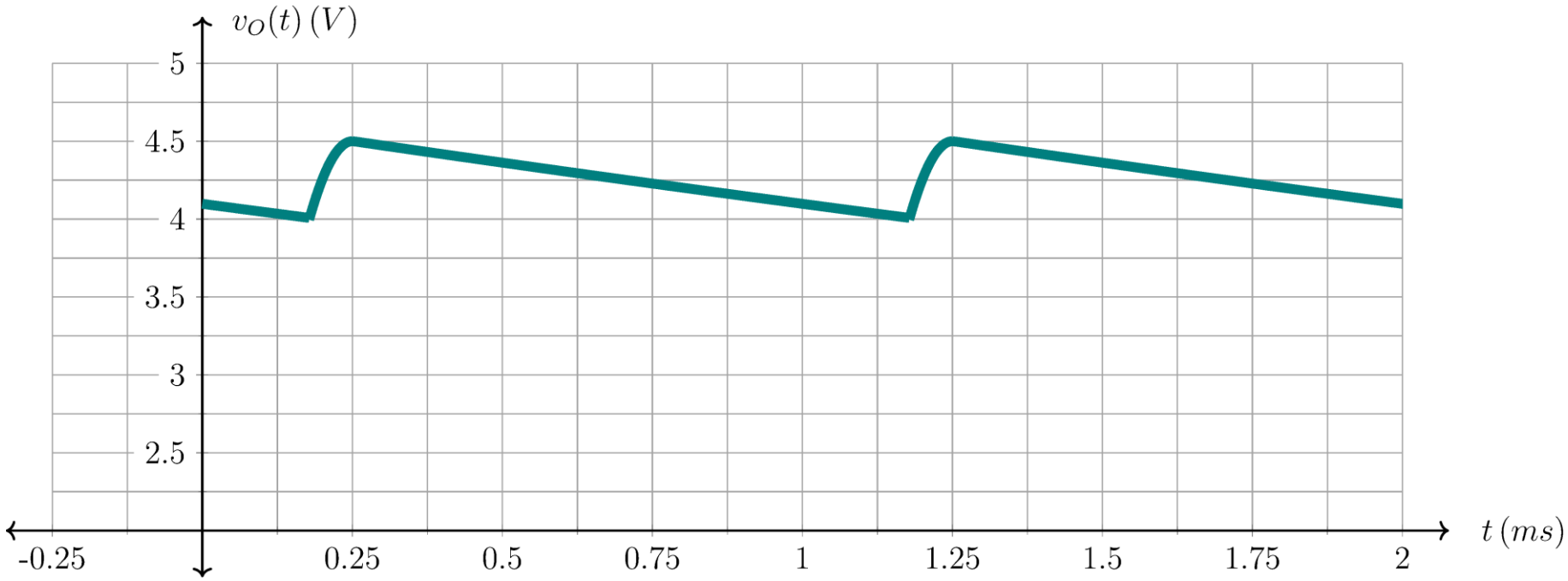
Problem 19

- The following is the input voltage to a full wave bridge rectifier without filter capacitor. Draw the output waveform on the same grid. Assume all the diodes are identical with a cut-in voltage $V_{D_o} = 0.5\,V$.



Problem 20

- The following is the rectified output of an ac sinusoidal voltage at $1\,kHz$ frequency, provided to a load of $4.5\,k\Omega$. The diode(s) used in the rectifier had a cut-in voltage of $0.5\,V$.
  I. What type of rectifier was utilized?
  II. What capacitance was used in the rectifier's design?
  III. What was the amplitude of the sinusoidal ac voltage applied to the rectifier?

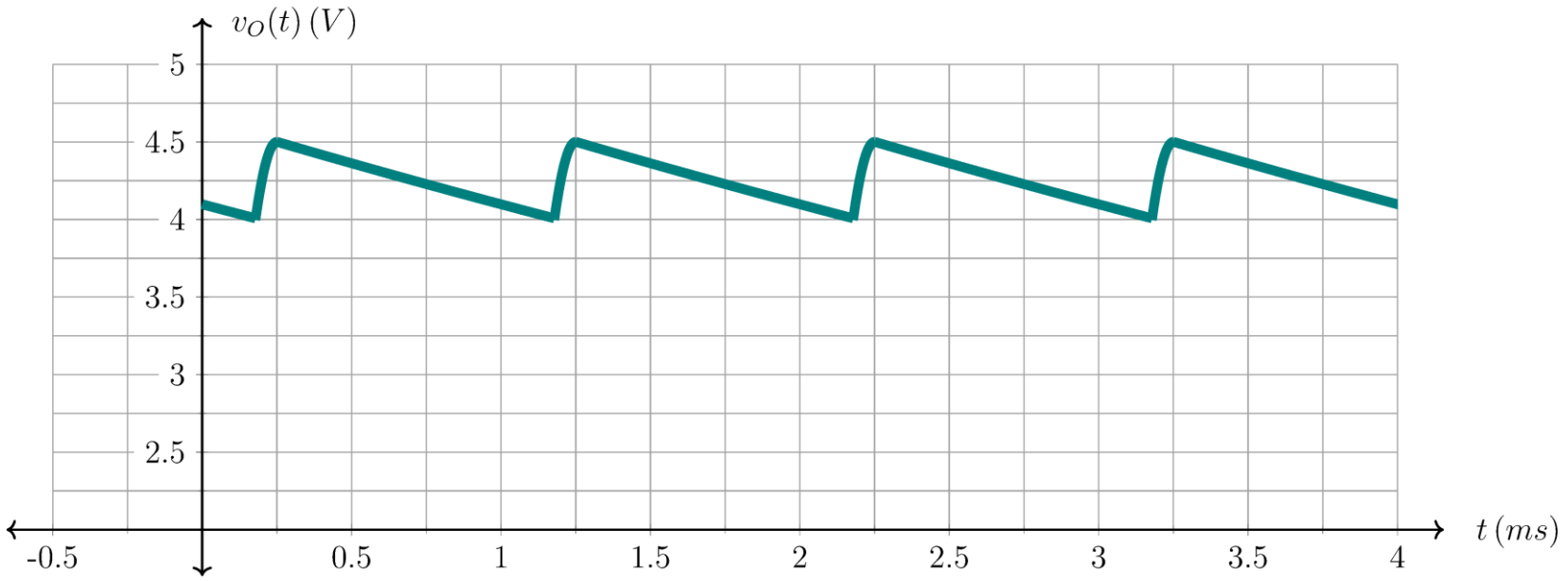


Ans: II. **2 μF**; III. **5 V**

Problem 21

- The following is the rectified output of an ac sinusoidal voltage at $500\,Hz$ frequency, provided to a load of $18\,k\Omega$. The diode(s) used in the rectifier had a cut-in voltage of $0.5\,V$.
  I. What type of rectifier was utilized?
  II. What capacitance was used in the rectifier's design?
  III. Write an expression as a function of time for the input voltage?
  IV. Draw the circuit diagram of the rectifier.

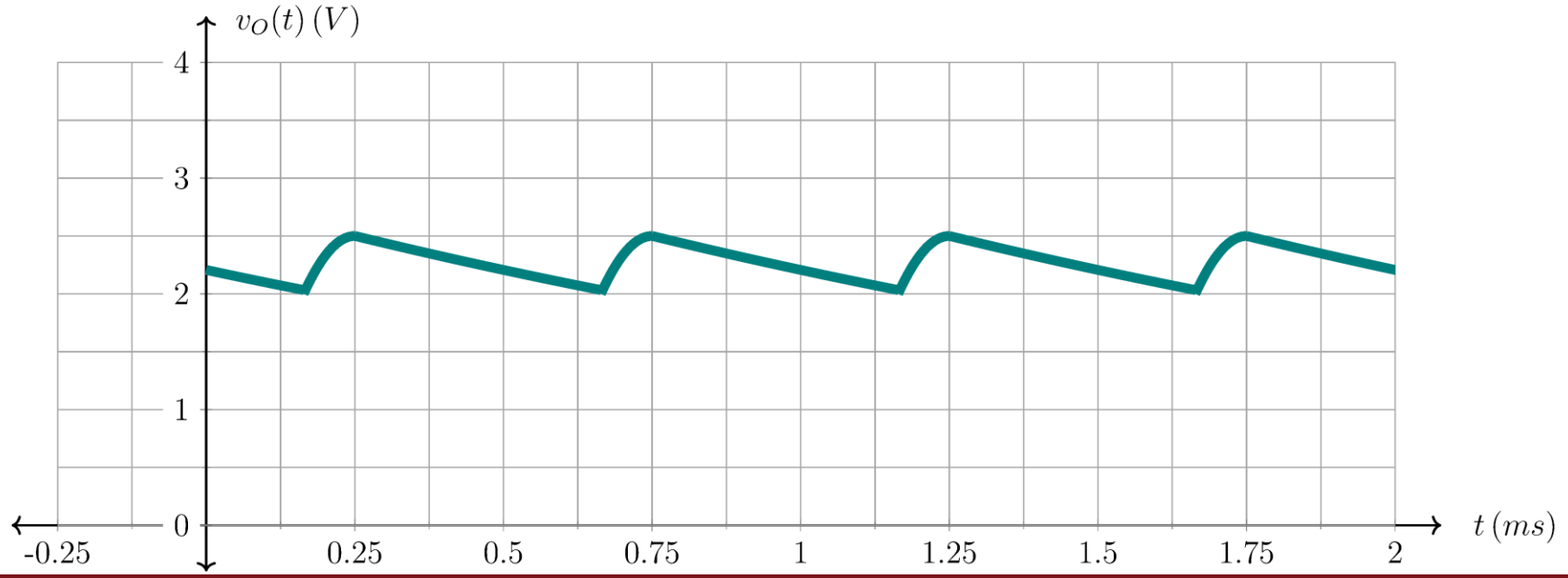
Ans: II. $\mathbf{0.5\,\mu F}$; III. $\frac{11}{2}\boldsymbol{sin}(\mathbf{1000\pi t})\,\boldsymbol{V}$



Problem 22

- The following is the rectified output of an ac sinusoidal voltage at $1\ kHz$ frequency, provided to a load of $5\ k\Omega$. The diode(s) used in the rectifier had a cut-in voltage of $0.5\ V$.
  I. What type of rectifier was utilized?
  II. What capacitance was used in the rectifier's design?
  III. Write an expression as a function of time for the input voltage?
  IV. Draw the circuit diagram of the rectifier.

Ans: II. $\mathbf{0.5\ \mu F}$; III. $\mathbf{\frac{7}{2}sin(2000\pi t)\ V}$



Up Next

Relevant *Questions* from Past Semesters

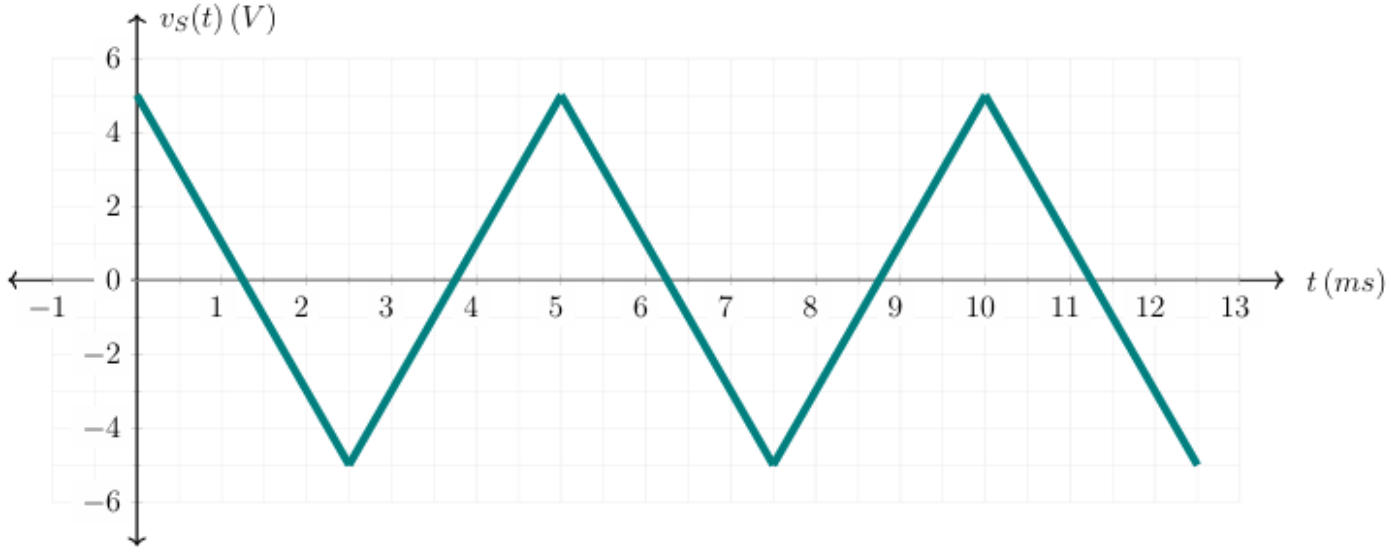
Final – Fall 2025

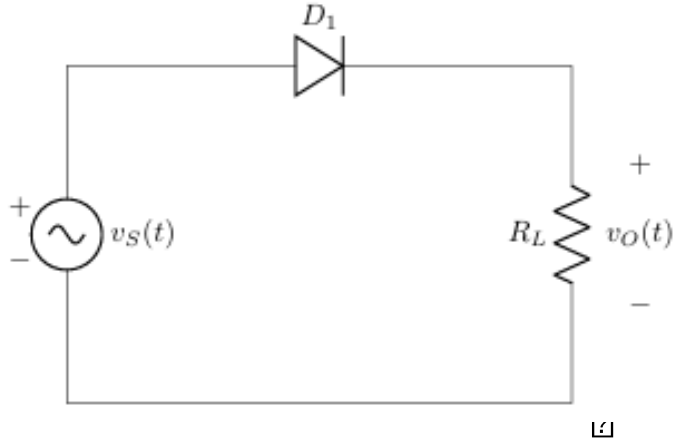
a. A $12\ V$ amplitude, $2\ kHz$ sinusoidal AC voltage is rectified and filtered using a FW rectifier with $4\ \mu F$ filter capacitor. The diodes used in the rectifier are identical and have a cut-in voltage of $1.2\ V$. If the average output voltage is measured to be $9\ V$, determine the value of the load resistance.

b. A rectifier circuit is designed to drive a resistive load R_L using a diode D_1 with cut-in voltages of $1\ V$, as shown in the schematic below. The circuit rectifies an AC input voltage $v_S(t)$, illustrated in the accompanying plot.

   i. On the same grid as $v_S(t)$, sketch approximately the waveform of the output voltage $v_O(t)$.

   ii. Draw the voltage transfer characteristic (VTC) graph for the rectifier on the following blank grid.





Ans: $(a)\ \boldsymbol{R_L = 500}$

Final – Summer 2025

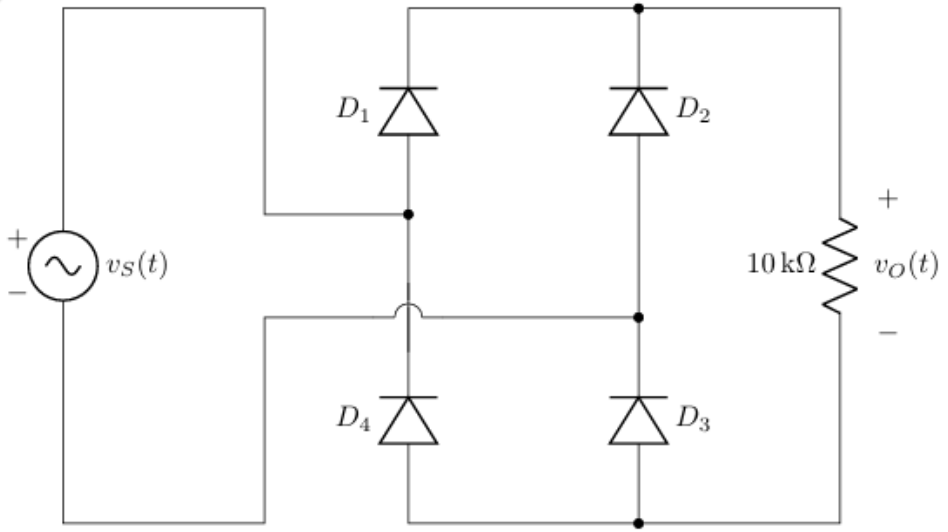
- A rectifier circuit is designed to drive a load of $10\ k\Omega$ using four diodes D_1, D_2, D_3, and D_4 with cut-in voltages of $0.5\ V$, $0.7\ V$, $0.5\ V$, and $0.3\ V$, respectively, as shown below. The circuit rectifies a sinusoidal AC input voltage $v_S(t)$, illustrated in the accompanying plot.

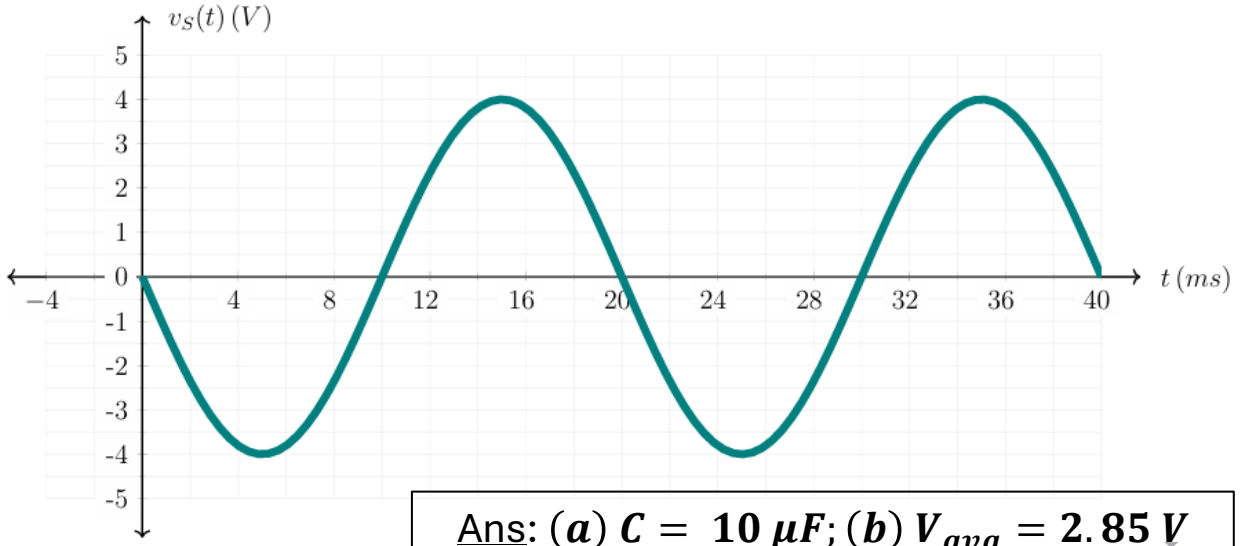
  a. On the same grid as $v_S(t)$, sketch approximately the waveform of the output voltage $v_O(t)$.

  b. A capacitor is connected in parallel with the load to filter the rectified voltage. If the output voltage fluctuation is restricted to 10% of its maximum value, determine the required capacitance.

  c. Determine the average output voltage corresponding to the filtered output in (b).

  d. Draw the voltage transfer characteristic (VTC) graph for the rectifier, that is, the plot of v_O versus v_S.

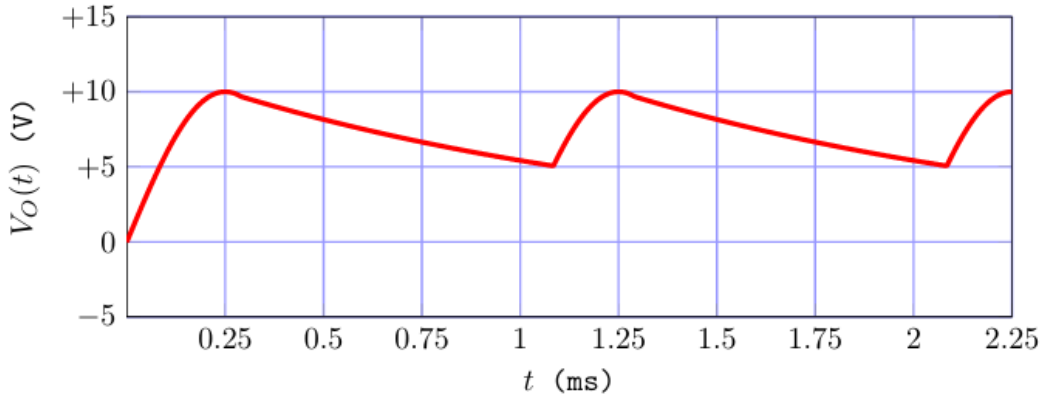




Ans: $(a)\ \boldsymbol{C = 10\ \mu F}$; $(b)\ \boldsymbol{V_{avg} = 2.85\ V}$

Final – Spring 2025

a. The adjacent graph is the output voltage waveform of a rectifier circuit. Calculate the output frequency and the average output voltage.



b. Design a rectifier circuit according to the specifications given below. Determine the appropriate values of the rectifier components and draw the circuit diagram of the rectifier designed. Here, V_{in} is the input voltage, f_{out} is the frequency of the output voltage, $V_{r(p-p)}$ is the peak-to-peak ripple voltage, and $V_{out(max)}$ is the maximum value of the output voltage. Assume $0.7\ V$ cut-in voltage for any diode.

$$V_{in} = 10\sin(200\pi t)\ V$$

$$f_{out} = 200\ Hz$$

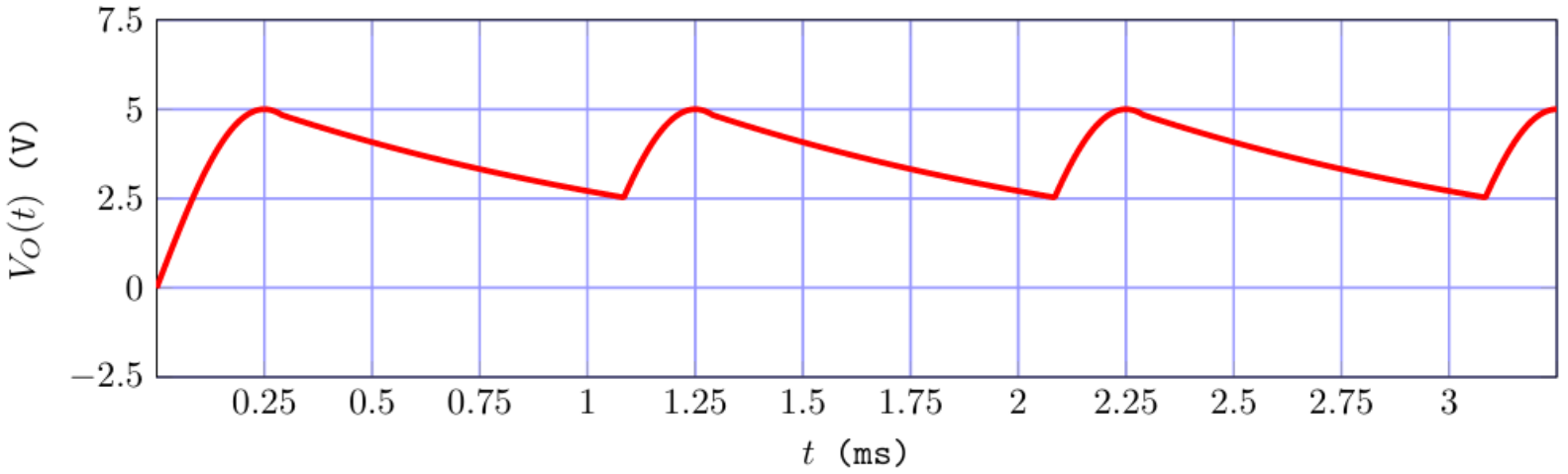
$$V_{r(p-p)} = 5\%\ of\ V_{out(max)}$$

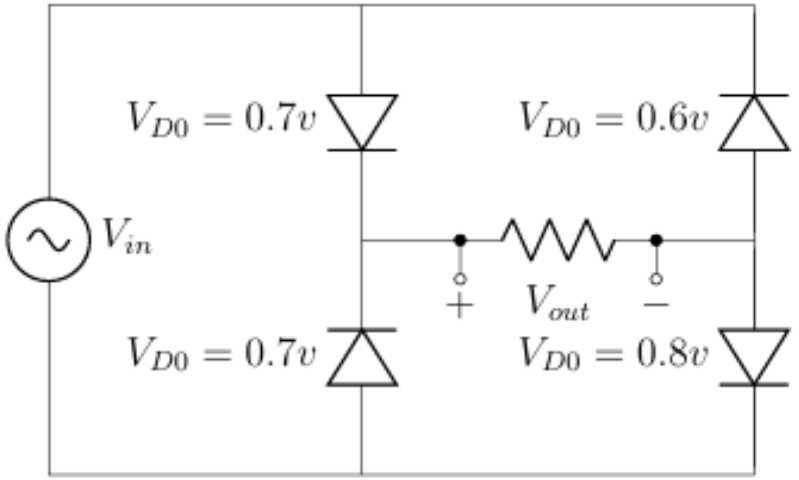
Ans: $(\boldsymbol{a})\ \boldsymbol{f_{out} = 1\ kHz,\ V_{avg} = V_{dc} \approx 7.5\ V;\ (b)\ RC = 0.1\ \Omega F}$

Final – Fall 2024 - 1/2

a. For the rectifier circuit shown below, draw the output voltage waveform for $V_{in} = 5\sin(100\pi t)\ V$.

b. The following is the plot of the output voltage of an unknown rectifier circuit with an output load resistance, $R = 5\ k\Omega$, input frequency, $f_{in} = 1\ kHz$, and cut-in voltages for the diode(s), $V_{D_0} = 0.5\ V$. Design the rectifier circuit, that is, determine the appropriate values of the rectifier components and draw the circuit diagram.



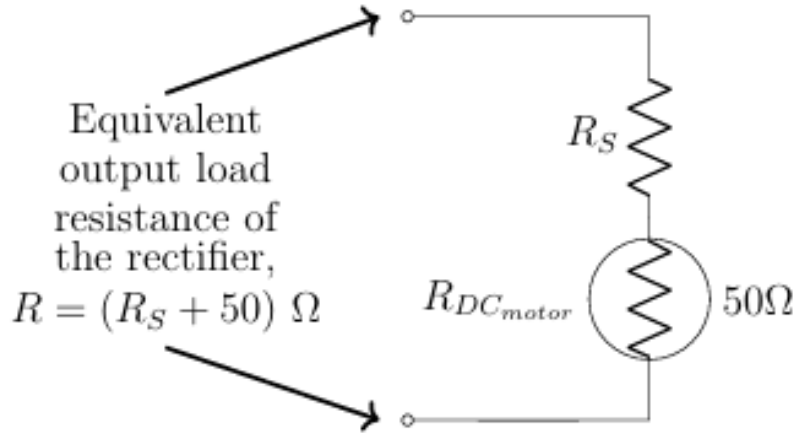


Ans: (b) ***HW Rectifier***, $\boldsymbol{C = 0.4\ \mu F}$

Final – Fall 2024 - 2/2

Shadman has access to a sinusoidal voltage source of, $V_{in} = 12\sin(500\pi t)\ V$ and he needs a DC voltage source to run a DC motor. The DC motor requires a very good quality DC voltage source to run. But it is difficult for him to get a pure DC voltage source. So, Shadman decided to build a rectifier that will take V_{in} as input and the output of the rectifier will work as the DC voltage source for the DC motor. While building the rectifier Shadman realized that, in order to make the DC motor run smoothly, the peak-to-peak ripple voltage of the rectifier should not exceed 2% of the peak input voltage. The DC motor has an internal resistance, $R_{DC_{motor}} = 50\ \Omega$. It is connected in series with a resistance, R_S to run smoothly, giving a total output load resistance, $R = (R_S + 50)\ \Omega$ for the rectifier circuit as shown in the following figure. Additionally, Shadman is building the rectifier circuit with diodes having a forward voltage drop of $V_{D_0} = 0.7\ V$.

Help Shadman design the rectifier circuit that meets the specifications mentioned above and determine the appropriate values of the rectifier components. Draw the designed circuit.



Mid – Spring 2024

- The input voltage of a half-wave rectifier is a Sine wave with amplitude $V_M = 10\,V$ and $40\,Hz$ frequency. The output load resistance is $R = 50\,k\Omega$. A silicon diode is used in this circuit whose forward voltage drop is $V_{D_0} = 0.6\,V$.

  a. Draw the Voltage Transfer Characteristic (VTC) curve of the rectifier.

  b. Calculate the DC value of the output voltage, V_{DC}.

  c. After connecting a capacitor parallel to the load, V_{DC} changes to $V_{DC(Cap)}$ and the new output voltage can be expressed as $V_{out} = V_{DC(Cap)} \pm 0.5\,V$. Calculate the peak-to-peak ripple voltage $V_{r(p-p)}$, and determine the ripple frequency, f_r. Now, calculate the value of the capacitor, C.

  d. Calculate the DC value of the output voltage after connecting the capacitor, $V_{DC(Cap)}$.

Ans: $\mathbf{(b)}\ \boldsymbol{V_{DC} = 2.88\,V};\ \boldsymbol{(c)}\ \boldsymbol{V_{r(p-p)} = 1\,V},\ \boldsymbol{f_r = 40\,Hz},\ \boldsymbol{C = 4.8\,\mu F};\ \boldsymbol{(d)}\ \boldsymbol{V_{DC(Cap)} = 9.1\,V}$

Mid – Summer 2023

- A voltage waveform $v_i = 5\sin(200\pi t)\,V$ is fed into a Full-wave rectifier with a load resistor, $R = 5\,k\Omega$. Silicon diodes are used in this circuit where, $V_{D_0} = 0.6\,V$.
  a. Calculate the DC value of the output voltage, V_{DC}.
  b. Draw the Voltage Transfer Characteristic (VTC) curve of the rectifier.
  c. Now, you have to connect a capacitor in parallel with the load resistor. You have two capacitors of $5\ \mu F$ and $1\ \mu F$ at your disposal. Which capacitor will you use? Explain briefly with necessary calculations.
  d. A different input waveform is fed into the Full-wave rectifier. The new peak-to-peak ripple voltage is 50% of the previous one calculated from (c) for the $5\ \mu F$ capacitor. The new output frequency is $300\,Hz$. Determine an expression of the input waveform.

Ans: $\mathbf{(a)}\ \boldsymbol{V_{DC} = 1.29\,V};\ \boldsymbol{(c)}\ V_{r(p-p),\,1\mu F} = 3.8\,V,\ V_{r(p-p),\,5\mu F} = 0.76\,V;\ \boldsymbol{(d)}\ v_i = 4.05\,sin(300\pi t)\,V$

Mid – Spring 2023

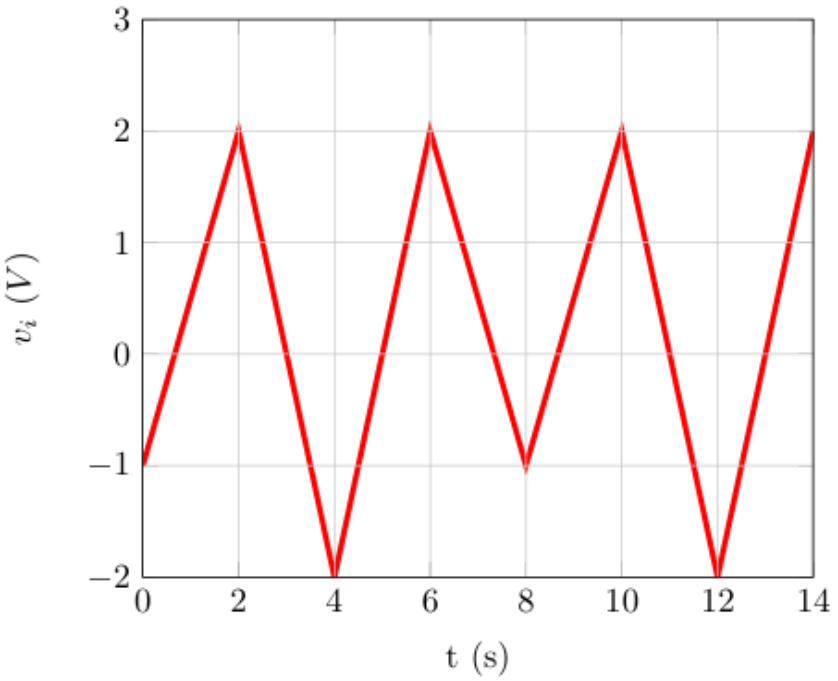
- A voltage waveform $v_i = 10\sin(1000\pi t)\,V$ is fed into a Full-wave rectifier with a load resistor, $R = 10\,k\Omega$. A capacitor is also connected in parallel with the load to reduce the fluctuation of the output voltage. It produces a peak-to-peak ripple voltage which is 3% of the peak output voltage. Silicon diodes are used in this circuit where, $V_{D_0} = 0.8\,V$.

  a. Deduce the peak output voltage, V_p, and the peak-to-peak ripple voltage, $v_{r(p-p)}$.

  b. Calculate the average (DC) value of the output voltage.

  c. Estimate the value of the capacitor from the given data.

  d. Now the input voltage v_i is changed to the one shown in the following figure and the capacitor is removed. Show the output waveform and indicate the voltage levels properly.



Ans: $(\mathbf{a})\ \boldsymbol{V_P = 8.4\,V},\ \boldsymbol{V_{r(p-p)} = 0.252\,V};\ (\boldsymbol{b})\ \boldsymbol{V_{DC} = 8.274\,V};\ (\boldsymbol{c})\ \boldsymbol{C = 3.33\,\mu F}$

Mid – Fall 2022

- A voltage waveform $v_i = 8\sin(100\pi t)\,V$ is fed into a rectifier with a load resistance $R = 10\,k\Omega$. Silicon diodes are used in this circuit for which the forward drop is $V_{D_0} = 0.8\,V$. The output frequency of the rectifier is $100\,Hz$.

  a. Identify the type of rectifier used (full-wave or half-wave). Give proper reasoning.

  b. Show the circuit with proper input and output labels.

  c. Illustrate the input and output waveforms in separate graphs. Label the graph and indicate the voltage levels properly.

  d. Calculate the DC value of the output.

  A capacitor is now added to reduce the fluctuation of the output voltage, which makes the peak-to-peak ripple voltage 5% of the maximum output voltage V_P.

  e. Deduce is the value of the capacitor from the given data.

  f. Calculate the DC value of the output after adding the capacitor and compare the result with that found in part (d).

Ans: $\mathbf{(a)\ FW\ Rectifier;\ (d)\ V_{DC} \approx 3.5\,V;\ (e)\ C = 20\,\mu F}$

Mid – Summer 2022

- The input voltage of a half-wave rectifier is a sinusoidal voltage with peak $V_M = 10\ V$ and $60\ Hz$ frequency. The output load resistance is $R = 2\ k\Omega$. A silicon diode is used in this circuit whose forward voltage drop is $V_{D_0} = 0.7\ V$.

  a. Approximately, sketch the input and the output waveforms.

  b. Calculate the DC value of the output voltage, V_{DC}.

  c. After connecting a capacitor parallel to the load, the output becomes $V_{out} = V_{DC(Cap)} \pm 0.2\ V$, where $V_{DC(Cap)}$ is the DC value of the output voltage after connecting the capacitor. Calculate the peak-to-peak ripple voltage $V_{r(p-p)}$ and the value of the capacitor, C.

  d. Calculate the DC value of the output voltage after connecting the capacitor, $V_{DC(Cap)}$.

Ans: $\mathbf{(b)}\ \boldsymbol{V_{DC} = 2.83\ V}$; $\boldsymbol{(c)\ V_{r(p-p)} = 0.4\ V,\ C = 193.75\ \mu F}$; $\boldsymbol{(d)\ V_{DC(Cap)} = 9.1\ V}$

Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
2. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill

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